Operations Research.

- The American Research Committee: It's Concerned with scientifically deciding how to best design and operate men-machine System, usually under ~~Conflict~~ Conditions requiring allocation of scarce resources.
- Operation Research can be seen as the use of science in solving real life problems with the aim of making the best out of scarce or limited resources.
- Operation Research is a scientific study of large system with a view to identify problem area and provide the managers with quantitative basis for decision which will enhance their effectiveness in achieving their Specified goal/objective.
- Operation Research is application of scientific approach to the effectiveness and efficient allocation and utilization of scarce resources to achieve their predetermined objective (goal) of such organization.

This implies being able to take optimal resource allocation decision in real management problems.

Operation Research Approach to Solving Problems

1. Observation and Identification of the problem that exist in the system.
2. Definition of the problem: A problem that is not properly defined cannot be easily solved.
3. Model formation: Model is the main body of operation Research that involves the Construction and the use of it. Model is a Simplified representation of a real life system which may still be an idea yet to be actualized.

Types of Model

1. Iconic Model
2. Analogue Model
3. Symbolic Model

Classical methods of Optimization

Optimization: The fundamental problem of optimization is to arrive at the best possible decision in any given set of circumstance.

Uses of classical optimization.

1. Classical optimization methods are used to obtain an optimal solution to certain kind of problems that involves continous and Differential functions.
2. We use it to obtain an optimal solution point of extreme of a certain type of problem that involves continous and differential functions.
3. They are analytical in nature and uses differential calculus in order to find the optimal value of the both constraint and unconstraints objective function.

The Maxima and Minima.

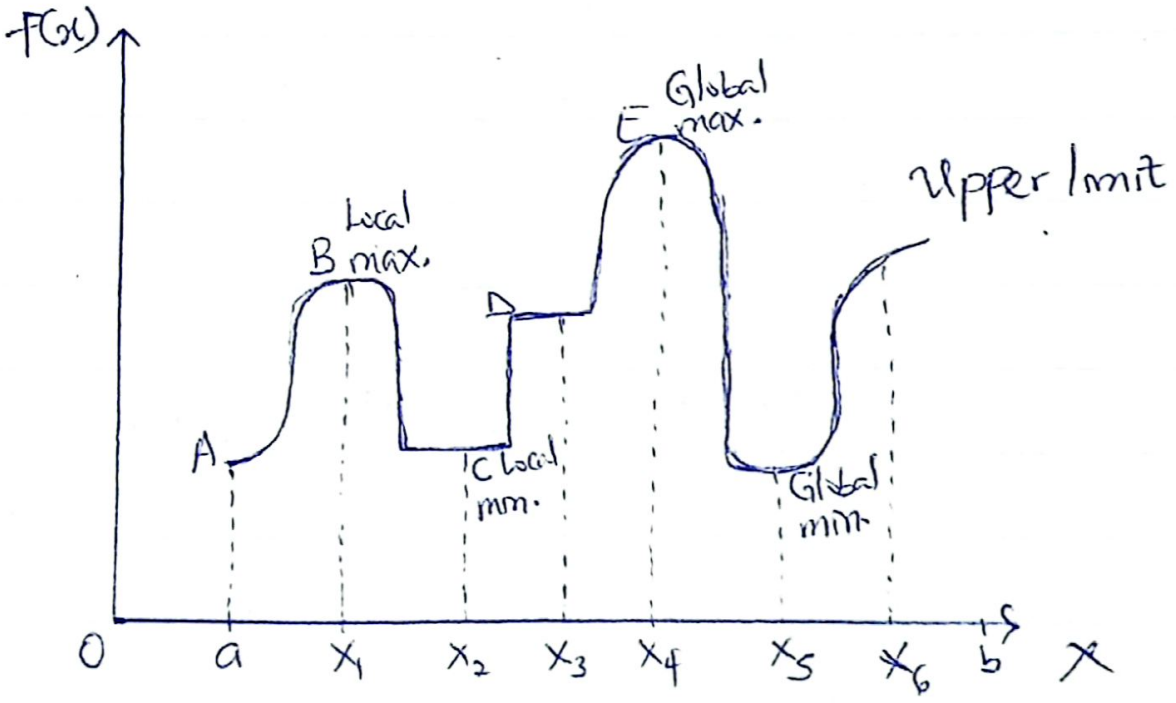
The maxima and minima are extreme points.

The necessary and Sufficient Conditions of optimization:

1. The problems with Unconstraints, Simple and multiple variables.
2. The problems with Constraints multivariable with equality and Inequality Constraints.

Unconstraints optimization: This can be defined mathematically, $y = f(x)$, the function $y = f(x)$ is said to be a maximum value at a point; $x = x_0$, if $f(x_0 + h) - f(x_0) < 0$ or $f(x_0) > f(x_0 + h)$; where h is sufficiently small number in the neighbourhood of point $x = x_0$. The point x_0 is a local maximum, if the value of $f(x)$ at every point in the neighbourhood of x_0 does not exceed $f(x_0)$, $f(x)$ is said to be a minimum value at a point $x = x_0$; If $f(x_0 + h) - f(x_0) > 0$ or $f(x_0) < f(x_0 + h)$. x_0 is called a strong global or absolute maximum if $f(x_0)$ is the highest among all other maximum points

Otherwise, it is called a local or relative maximum. X_0 is called a strong global or absolute minimum. If $F(X_0)$ is the highest among all other maximum points otherwise, it is called a local or relative maximum. X_0 is called a strong global or absolute minimum. If $f(X_0)$ is the highest among all other minimum points, otherwise it is called a local or relative minimum.



1) The domain is the range of values of x
2) The domain limit [end points] are generally

Called stationary points or Critical points.

3. There are two Categories of stationality
   1) The Inflection point
   2) The extreme point [Global or absolute]
                        [local or relative]

$x_1, x_2, x_3, x_4, x_5, x_6$ are all extreme points of $f(x)$. x_1, x_4, x_6 are the maximum points. Since $f(x) = \max f(x) f(x_y) f(x_6)$. x_5 is the global or absolute minimum point Since $f(x_5) = \text{Max} f(x_2), f(x_3), f(x_5)$.

x_3 is the Saddle point or Inflection. x_1 is the local maximum and x_2 is the local minimum.

Necessary Conditions for local minimum and maximum.

* For a point x_0 to be local extreme of a function $y = f(x)$ defined in the Intervals $a \leq x \leq b$ is that the first derivative of $f(x)$ exist as a finite

number at $x = x_0$ and $f'(x) = 0$.

* If the sign of $f'(x_0)$ changes from positive to negative as x increases in the neighbourhood of $x = x_0$, the value ~~of~~ * $f(x)$ will be local maximum.

* If the sign of $f'(x_0)$ changes from negative to positive as x increases in the neighbourhood of $x = x_0$, the value of $f(x)$ will be local minimum.

* The necessary condition for any function $f(x)$ to have a local optimum value at any extreme point of $x = x_0$, is that its first derivative, $f'(x_0) = 0$ or $\nabla f(x_0) = 0$ is called Stationary point.

SUFFICIENT CONDITION

If at any extreme point of $X = X_0$ of $f(x)$, the first $(n-1)$ derivative of it becomes 0. i.e $f'(x_0) = f''(x_0) = f(x_0)^{n-1} = 0$; $f^n(x_0) \neq 0$

Case 1: If $f''(x_0) > 0$, then both $f(x_0+h) - f(x_0)$ are positive, hence local minimum value of $f(x)$ exist at $x = x_0$.

Case 2: If $f''(x_0) < 0$, the both $f(x_0+h) - f(x_0)$ are negative and hence local maximum values of $f(x)$ exist at $x = x_0$.

Case 3: If $f''(x_0) = 0$, then no information is obtained about the maximum or minimum value of $f(x)$, it is point of Inflection. That is:

1) Positive definite (> 0) when x_0 is a minimum point.
2) Negative definite (< 0) when x_0 is a maximum point.
3) Indefinite ($= 0$) when x_0 is a saddle point.

Rule for determining maxima and minima in one variable case:

Suppose we want to optimise $z=f(x)$

Firstly, we obtain the first derivative, $z' = dz/dx = f(x)$ --- (i) equate to 0 and solve the equation

$$f(x) = 0 \quad \text{---} \quad \text{(ii)}$$

Secondly, obtain the second derivative of z; $z'' = d^2z/dx = f''(x)$ --- (iii)

Substitute the values of x obtained in equation (ii) into equation (iii) and examine the sign of $f''(x)$. If:

a) $f''(x) > 0$, it is positive, the $f(x)$ has a minimum value.

b) If $f''(x) < 0$, it is negative, the $f(x)$ has a maximum value.

c) If $f''(x) = 0$, then x is a saddle point.

Example: Find the maxima or the minima of the function:

$$z = 2x^2 - 4x + 1$$

$$z' = 4x - 4$$

$$4x - 4 = 0 \; ; \; x = 1$$

$z'' = 4 \quad ; \; 4 > 0$.

Example: Find the optimum value

$z = x^3 - 6x^2 + 9x + 1$

$z' = 3x^2 - 12x + 9$

$3x^2 - 12x + 9 = 0$

The factors $x = 3, \; x = 1$

$z'' = 6x - 12$

$6x - 12 = 0 \; ; \; x = 2$

$x = 1 : \; 6(1) - 12 = -6$ (max)

$x = 3 : \; 6(3) - 12 = 6$ (min)

Using $-6 < 0$

Equate $x = 1, 3$ in $x^3 - 6x^2 + 9x + 1$

when $x = 3 : (3)^3 - 6(3)^2 + 9(3) + 1$

$27 - 54 + 27 + 1 = 1$ (min value)

when $x = 1 : (1)^2 - 6(1)^2 + 9(1) + 1$

$1 - 6 + 9 + 1 = 5$ (max. value)

A firm has a total revenue of $r = 20x - 2x^2$ and a Total Cost function $C = x^2 - 4x + 20$ where x represent the quantity. Find the revenue maximum output level and the corresponding value of

profit price and total revenue.

Sol.

The necessary condition for revenue function, r is to have a maximum value. i.e: $\frac{\partial R}{\partial x} = 0$ and $\frac{\partial^2 R}{\partial x^2} < 0$

Since the value of the 2nd derivative is negative (−ve), the revenue level will be maximum.

Profit $C = (20x - 2x^2) - (x^2 - 4x + 20)$

$= 20x - 2x^2 - x^2 + 4x - 20$

$= 24x - 3x^2 - 20$

Substitute $x = 5$

$= 24(5) - 3(5)^2 - 20$

$= 120 - 75 - 20 = 25$

Case of Two variables:

If $Z = f(x, y)$

1) Firstly, take the partial derivative of f w.r.t.o x and y respectively. Equate to zero and solve for x and y.

$\partial f/\partial x = 0$, $\partial f/\partial y = 0$ ——(1)

2) Find the second order partial deriv of u that is:

$$\partial^2 u/\partial x = \partial^2 u/\partial x \cdot \partial^2 u/\partial y$$

If $\frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} - \left(\frac{\partial^2 u}{\partial x \partial y}\right)^2 > 0$ and

$\frac{\partial^2 u}{\partial x} > 0$, $\frac{\partial^2 u}{\partial y} > 0$ then $u(x,y)$ has a minimum value ~~of~~ at (x,y) as obtained in equation (1).

If: $\frac{\partial^2 u}{\partial x} \cdot \frac{\partial^2 u}{\partial y} - \left(\frac{\partial^2 u}{\partial x \partial y}\right)^2 > 0$ and

$\frac{\partial^2 u}{\partial x} < 0$, $\frac{\partial^2 u}{\partial y} < 0$, Then $u(x,y)$ has a maximum value $u(x,y)$ obtained from eqn 1

Example 1: Find the maximum value if

$f(x,y) = z = 2x^2y(1-x-y)$

$$z = 2x^2y - 2x^3y - 2x^2y^2$$

$$\frac{\partial z}{\partial x} = 4xy - 6x^2y - 4xy^2 \quad --(1)$$

$$\frac{\partial z}{\partial y} = 2x^2 - 2x^3 - 4x^2y \quad --(2)$$

$$4xy - 6x^2y - 4xy^2$$

$$2xy(2-3x-2y)=0$$

$$2xy=0$$

$$2-3x-2y=0$$

$$-3x-2y=-2$$

$$3x+2y=2 \quad --(1)$$

$$2x^2-2x^3-4x^2y^2$$

$$2x^2(1-3x-2y)=0$$

$$2x^2=0$$

$$1-x-2y=0$$

$$-x-2y=-1$$

$$x+2y=1 \quad ---(2)$$

$$3x+2y=2$$

$$-\ x+2y=1$$

$$2x=1 \;;\; x=1/2 \text{ //}$$

Sub. $1/2$ for x in $3x+2y=2$; $3(\frac{1}{2})+2y=2$

$$\frac{3}{2}+2y=2 \;;\; 2y=\frac{2}{1}-\frac{3}{2}$$

$$\frac{1}{2}\times 2y=\frac{4-3}{2}=\frac{1}{2}\times\frac{1}{2}$$

$$y=\frac{1}{4} \text{ //}$$

$$x=1/2,\ y=1/4 \text{ //}$$

$$\frac{\partial^2 u}{\partial x} = 4y - 12xy - 4y^2 = -3/4 < 0$$

$$\frac{\partial^2 u}{\partial y} = -4x^2 = -1 < 0$$

The point value when $x = 1/2$ and $y = 1/4$, the maximum is gotten:

$$z = 2x^2y - 2x^3y - 2x^2y^2 \neq$$

$$z = 2\left(\frac{1}{2}\right)^2\left(\frac{1}{4}\right) - 2\left(\frac{1}{2}\right)^3\left(\frac{1}{4}\right) - 2\left(\frac{1}{2}\right)^2\left(\frac{1}{4}\right)^2$$

$$z = 1/32 \text{ (The maximum value).}$$

Exercises:

1) $f(x) = 4x^4 - x^2 + 5$
2) $f(x) = 6x^5 - 4x^3 + 10$
3) $f(x) = x^3 - y^2 - 3xy$
4) $f(x) = x_1^2 + 5x_2^2 + 7x_1x_2$